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How to Improve Utility Performance: Understanding Structural, Governance and Regulatory Incentives in Kenya Power and Kenya

Peter Twesigye

Abstract: The power sectors in most African countries face an enduring problem of poor utility performance. Empirical findings show that the structural, governance, and regulatory reforms that previously created incentives for improved utility performance are increasingly threatened by political influence. Kenya Power's financial viability has deteriorated in recent years and the regulator has been undermined. One of my major conclusions is that when the relationship between the government and utility is well understood and the agent is properly incentivized, performance improvements are possible. However, when the principal undermines or muddies those incentives through conflicting political interventions, performance improvements can be reversed.

Keywords: power sector reforms, utility regulation, governance, energy access, performance evaluation, agency theory

JEL Classification Codes: G01, G21, G38, E44, H12

Kenya's history and experience with power sector reform is unique and different from other countries in the region thus affording the potential for interesting and valuable insights. Kenya was one of the few African countries that decided early on to unbundle generation from transmission and distribution, with the creation of Kenya Electricity Generation Company Limited (KenGen) and Kenya Power and Lighting Company Limited (KPLC[1]). Uniquely, both KPLC and KenGen are listed on the Nairobi Securities Exchange (NSE) although the government retained majority ownership (50.1% in KPLC and 70% in KenGen). For this reason, we refer to them as Mixed Capital Enterprises. One of the consequences of unbundling generation from transmission and distribution has been the removal of the potential for a conflict of interest[2] which arises where a vertically integrated, national utility is both a generator and the single buyer for Independent Power Producers (IPPs).

Supported by a national planning process for least-cost generation expansion, KPLC was able to translate plans into timely initiation of competitive bids for new power capacity

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[1] KPLC was rebranded as Kenya Power in 2011, but for uniformity in this paper, I refer to it as KPLC
[2] Competing in procuring new generation capacity and power dispatch to the grid

(Eberhard et al. 2016). Through building in-house capabilities, KPLC ran a series of successful tenders, mostly for thermal power, with subsequent tender rounds yielding even lower prices, and plants generally being built on time. Kenya ranks amongst the countries with the highest number of IPPs in sub-Saharan Africa and currently is in the fortunate position of having surplus capacity. However, these planning and procurement capabilities have eroded and dissipated somewhat in recent years. Kenya was also amongst the early countries to establish an independent electricity regulator, which later was converted into an energy sector regulator. In addition, the utility has built a strong institutional culture and technical capacity fostered through targeted capacity development and training and supported by involved leadership. The utility was subject to a private management contract for a short period. These reforms, especially the partial listing of KPLC, create an interesting set of governance and regulatory incentives for improved performance.

Despite these advancements, the power sector in Kenya still faces enduring challenges, including weaknesses in the transmission and distribution segments of the system, the politicization of long-term planning, an emerging surplus of power generation capacity, and the need to integrate more variable sources of energy without affecting system stability (KPLC 2018). In addition, recent gains in electricity access and generation capacity have come at a cost: threats to the financial sustainability of KPLC (Mussa 2021) and KenGen have necessitated periodic debt restructuring. KPLC's profitability, which for decades was steadfast, has, in recent years, been in decline, unable to meet its debt covenants, and its solvency and creditworthiness are increasingly threatened (Business Daily 2018; The Star 2020; KPLC 2019). Recent governance challenges saw the termination of the entire senior management team in 2018 and the board in 2020 (Muli 2020).

This article examines the experience of Kenya through the lens of power sector reform and principal-agent theory, yielding new insights, explanations, and knowledge about the relationship between power sector regulatory, structural, and governance reforms and how they impact utility performance—mainly increasing access and delivering adequate and reliable electricity at competitive prices through a utility which strives to be technically efficient, financially viable, creditworthy and able to attract investment into the sector. The article goes further in seeking to understand how these reforms alter governance and regulatory arrangements to drive performance.

I use an analytical framework that combines the power sector reform and the principal-agency theories which enables a deeper understanding of how strong governance and structural reforms have provided stronger incentives for improved performance.

Enduring Power Challenges in Sub-Saharan Africa in Brief

The Sub-Saharan Africa (SSA) region including Kenya is faced with five key enduring power challenges which have, in combination, constrained delivery of electricity services on a sustainable basis.

(i) Installed generation capacity is inadequate, with most countries in the region having power systems smaller than 500MW (IEA 2019b).
(ii) Electricity access rates remain low, with 45% of households not being connected to the grid (IEA 2019a; Blimpo and Postepska 2017).
(iii) Even for those with a connection, power cuts and load-shedding is a frequent occurrence due to inadequate generation capacity and inadequate investments and maintenance in the network (Kojima et al. 2016; Eberhard et al. 2016).

(iv) The cost of electricity (median tariff is US$0.15 per kilowatt hour) in Africa is among the highest in the world (Trimble et al. 2016; Huenteler et al. 2017).
(v) Households and enterprises often have to rely on expensive diesel back-up power generation to meet their electricity needs (Eberhard et al. 2016; Kojima et al. 2016; Andersen and Dalgaard 2013).
(vi) The above problems arise because of the poor financial performance of utilities. Technical inefficiencies are reflected in high losses and inefficiencies in capital expenditure (CapEx) execution (Eberhard 2020), which result in the high cost and poor quality of service (Kojima et al. 2016; Trimble et al. 2016).
(vii) Commercial inefficiencies are reflected in poor billing and collections, leading to chronic indebtedness (Eberhard and Dyson 2019), underpricing or below-cost tariffs, and poor customer service leading to low willingness to pay.

Consequently, most incumbent electricity distribution companies (DisCos) are financially distressed and dysfunctional. The resultant revenue gap imposes an additional burden on the already strained government fiscal resources, a trend that has persisted in most SSA countries.

A response to these challenges has been utilizing power sector reforms involving corporatization, regulation, restructuring, competition, and private sector participation (PSP).

Linking Power Sector Reforms to Utility Performance

Poor technical and financial performance was the defining feature of many African electricity sectors by the end of the 1980s and 1990s (Gratwick and Eberhard 2008; Williams and Ghanadan 2006). Drawing on the successful electricity sector liberalizations in the United States and England, the World Bank began to promote, including in Africa, significant structural changes in developing countries' electricity sectors alongside the broader structural adjustment programs for liberalization (Foster and Anshul 2019; Besant-Jones 2006). Over time, the reforms came to be known as the "standard model" and involved the establishment of an independent regulator, restructuring, PSP, and competition (Twesigye 2022).

Despite many countries initially committing to the "standard model," these reforms have progressed only partially, and differently, across the region, mainly resulting in hybrid power market structures, in which dominant incumbent state-owned utilities continue to operate alongside independent power producers (Gratwick and Eberhard 2008). Some countries have also incorporated private management contracts or long-term concessions (Eberhard et al. 2017). These hybrid market structures have resulted in new governance and regulatory frameworks and have generated new operational and commercial issues that have impacted performance. In addition, the reforms have sent different sets of signals around incentives in the structural, governance, and regulatory frameworks of utilities, unlike what was initially envisaged in the pioneering standard model reforms.

However, the impact of these reforms on utility performance has been mixed. Several studies have been conducted to establish the reform impacts on performance (for example: Jamasb et al. 2014; Polemis 2016; Urpelainen, Yang, and Liu 2017; Zhang, Parker, and Kirkpatrick 2008). However, these have mainly been narrow in scope and use econometric/statistical approaches to explore relationships between a limited number of variables. Yet, the design of power sector reforms involves a range of interventions that play out in different ways depending on the political economy and country's context; these should be adequately taken into account.

This article provides a deeper exploration of how the core reform steps of regulation, restructuring, competition, and PSP impact utility performance in Kenya to answer the following questions:

(1) How can we explain and understand the varied utility performance outcomes in Kenya?
(2) To what extent does principal-agent theory, combined with power sector reform theory provide an analytical framework to explain the varied performance?
(3) In which ways do the power sector reforms alter principal-agent relationships and incentivize improvements in performance in Kenya?

The article continues with the presentation of the materials and methods, followed by a description of the theoretical and analytical framework. The analysis of reforms, governance arrangements and operational performance provides quantitative and qualitative results and discussion. The conclusions and policy implications are presented last.

Materials and Methods

The research utilizes a qualitative explanatory case study with embedded mixed methods of data collection (quantitative and qualitative) and analysis (Saunders, Lewis, and Thornhil 2019) to investigate drivers of utility performance, governance arrangements, and principal-agency relationships. Cases are good at revealing the complete picture considering the context (Eisenhardt 1991; Yin 1994), and identifying causal links/operational pathways through the rich and in-depth information, firmly rooted in the original evidence (Langley 1999). This case offers deeper insights into structural, governance, and incentive arrangements within utilities, and can explain potential causal principal-agent pathways of institutional and organizational changes that are difficult to identify with econometric and statistical models (George and Bennett 1997).

Data Collection and Analysis

Two methods of data collection were embedded within the case studies: quantitative data collection was conducted first from secondary and primary data sources, including annual reports, sector-wide data, and multilateral development agency databases. The data was analyzed manually using Excel spreadsheets.

The qualitative phase involved qualitative document analysis (QDA) of secondary data (Bowen 2009). The documents included contracts and agreements, regulations and licenses, sector policies, annual reports, management reports backed by primary data, utility policy documents and board charters, company fact sheets, energy sector publications and reports, archival records, observational data from meetings, as well as official media publications from utilities and regulatory agencies. Document analysis allowed for the identification of conceptual categories from the literature on power sector reforms and formulation of emerging themes of the most influential factors on utility performance and their causal relationships with the outcomes (measured as KPIs).

These relationships were further explored through interviews. A purposive sample of ten utility top executives and senior managers—high priority and highly knowledgeable interviewees who viewed the focal phenomenon of utility performance, governance, and incentives frameworks from diverse perspectives—were interviewed to obtain both retrospective and real-time accounts and reasons behind the utility performance phenomena, governance arrangements and principal-agency problems/relationships. Informed consent was obtained from interviewees prior to the interviews.

To limit information bias, a sample of two regulatory authority senior executives or managers helped validate the information utilities provided. Combining these methods was necessary to allow for a direct assessment and data validation. The qualitative data were analyzed using NVivo software and triangulated (Guba and Lincoln 1994).

Theoretical and Analytical Framework

The research is scoped first in the theories of economic regulation of infrastructure systems —both positive[3] and normative[4] (Robinson 2005; Brown et al. 2006). Economic regulation is necessary for four reasons: First, to enable the government (principal) to overcome information asymmetry with the utility (agent) and align the utility's interests with the government's interests (Newbery 2002; Robinson 2005; Berg 2019). Second, there is a need to protect customers from natural monopolistic market power when competition is non-existent or ineffective (Robinson 2005; Creti and Fontini 2019). Third, there is a need to protect utilities/operators from rivals, and last, utilities or operators also desire protection from government opportunism (Stigler 1971).

I, therefore, adopt the Principal-Agent theory to analyze and address issues of information asymmetry that exist between the utility and regulator, and how competition and incentive regulation can be used to limit the monopolistic market power (Sappington 1991). Competition helps to induce the agent to improve its performance by revealing its true operating costs (Newbery 2002) and align the utility's interests with those of the customers, government, and other private information. Competition also helps to contain monopoly and market power enabling regulators and government to revise the structure of the utility industry. In this framework, the government is the principal, and the utility is the agent whether publicly or privately owned. Rate of Return Regulation (RoR)[5] is applied in Kenya and is used in the analysis to give an understanding of price setting and resultant tariffs.

Using the positivist principal–agency[6] theoretical lens (Eisenhardt 1989), the analytical framework in figure 1 presents a governance structure that is used to explain the inter-relationships of various actors in the Kenyan electricity sector as well as the structural, governance and regulatory incentives on utility performance. The aim is to understand how the principal-agent theory, combined with power sector reform theories, provides a more powerful analytical framework to better explain the performances of utilities. Agency theory further helps us to explore the behavior and incentives accruing to principals and agents as motivation for their actions—using a metaphor of a contract. Bringing these two bodies of

[3] Positive theory of regulation examines why regulation occurs and assesses how monopolies obtain and exercise market power on resultant market structures. It also assesses the various interest groups in the industry and possible government opportunism and describes why restrictions on government discretion is necessary for the energy sector to provide efficient services for customers.

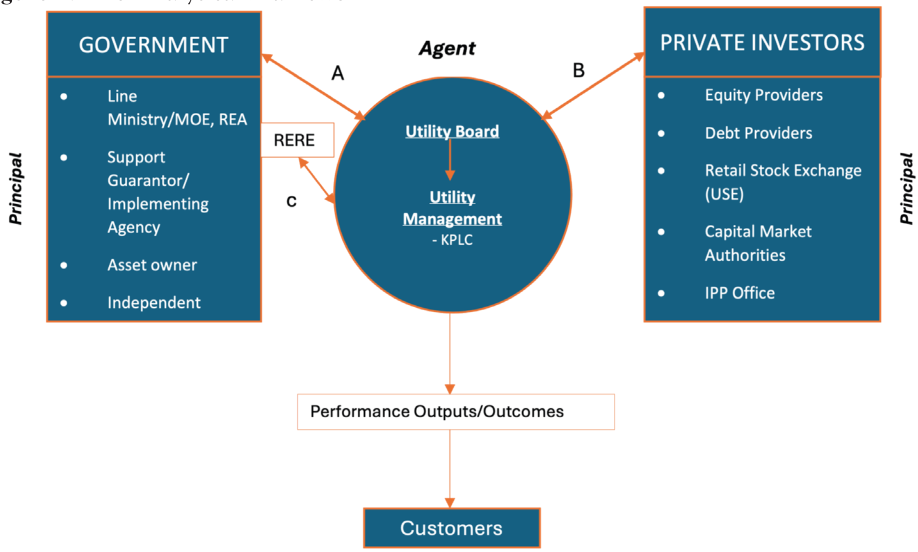
[4] Normative theory of regulation assesses how regulators should encourage competition in the market to minimize costs of information asymmetry by obtaining information and providing incentives to utilities/operators to improve their performance. It focuses on the design of price structures that improve economic efficiency and regulatory arrangements or rules that empower regulators to be independent, transparent, predictability, credibility and with legitimacy in decision making.

[5] Under this regulatory contract, the utility (KPLC, in this case) must justify its price by demonstrating that its return on capital does not exceed a specified target rate. ROR is the most used tariff methodology in Africa.

[6] The positivist school of thought is based on identifying situations in which the principal and agent have conflicting goals and then describing the contract or governance mechanisms that limit the agent's selfish behavior.

theory together provides a potentially powerful analytical framework for comparison and understanding of utility performance.

Figure 1. The Analytical Framework



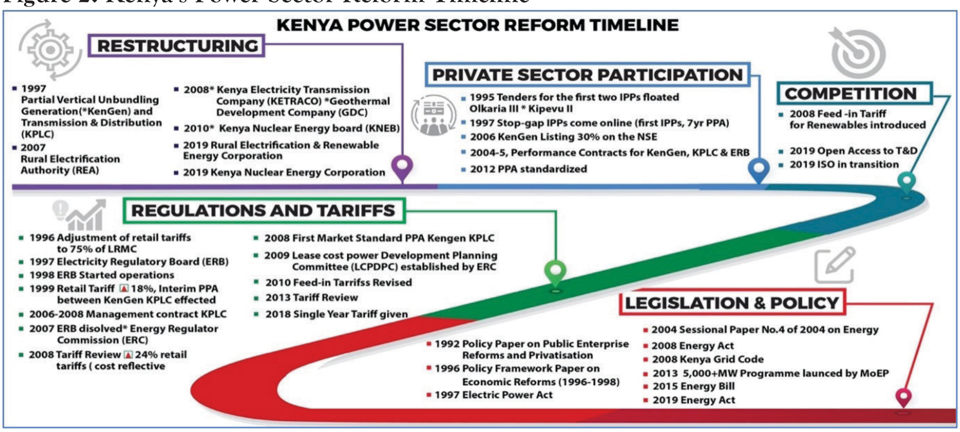
Source: Author's creation.

Analysis, Results, and Discussion

Power Sector Reforms in Kenya

Figure 2 shows the key reform components and timeline of reforms in Kenya's power sector.

Figure 2. Kenya's Power Sector Reform Timeline



Source: Author's creation.

During the period 1990 to 1994, Kenya experienced major macroeconomic challenges resulting from a confluence of factors, including drought and deterioration in its terms of trade. Donor funding to the power sector was frozen between 1991 and 1996, mainly because of governance failures linked to allegations of corruption. The 1992 policy paper on

Public Enterprise Reforms and Privatization was one of the responses to the aid embargo. It set the basis for later reforms in the power sector and deregulation of the Kenyan economy that began in 1993 (Godinho and Eberhard 2019). Over this period the IMF and the World Bank outlined terms for lending that were conditional on reforms, including tariff studies, reorganization of the power sector, and legal and regulatory reforms. Subsequently, a move towards cost-reflective tariffs[7] (a component of commercialization) was adopted.

As the country emerged from an aid embargo, one of the state's main objectives was to attract much-needed private sector investment to complement limited public sector investment. In a policy paper on economic reforms at the time (Government of Kenya 1996), the government outlined its intention to separate the regulatory and commercial functions of the sector, to facilitate restructuring, and to promote private-sector investment, including via IPPs (following the recommendations of the World Bank and the IMF). Meanwhile, the power situation deteriorated further. In the agreement reached at the end of 1996, World Bank funding was made conditional on unbundling the generation segment from KPLC, restructuring, and enacting enabling legislation. Consequently, the Electric Power Act[8] of 1997 was passed (Eberhard, Gratwick, and Kariuki 2018).

At the industry level, rationalization and unbundling redefined the scope[9] of KPLC, which had operated as an integrated utility since 1954.

In addition to restructuring and reorganizing regulatory functions, KPLC, with support of the World Bank and external consultants, began the first competitive process for tendering and negotiating IPP contracts. In 2003, the government expressed dissatisfaction with the performance of the energy sector (Government of Kenya 2003), noting that, despite the reforms, including the introduction of IPPs, electricity in Kenya was still unreliable and expensive. To remedy this, a deeper, second wave of reforms was recommended and subsequently detailed in the national energy policy of 2004, commonly known as Sessional Paper No. 4 of 2004 (Government of Kenya 2004), which set the basis for passage of the 2006 Energy Act and outlined the government's commitment to:

- establish a rural electrification authority;
- accelerate the increase in the rural electrification rate by 10 percent a year;
- facilitate the development of a competitive market structure for the generation, distribution, and supply of electricity;
- establish the Geothermal Development Company (GDC) to assess Kenya's geothermal resources, including steam-field appraisal and development;
- establish Kenya Electricity Transmission Company Ltd (KETRACO) to build new transmission lines using government funds and concessionary financing;
- provide an increase in the lifeline tariff for domestic consumers of up to 50 kWh per month from July 2008;
- enact new legislation to, among other things, dissolve the ERB and create a new independent energy sector regulator—the Energy Regulatory Commission10 (ERC), and

[7] In 1997, tariffs were increased to 75% of long-run marginal costs.

[8] The government's primary function, through the Ministry of Energy and Petroleum (MoEP), became policy formulation, and its regulatory authority was devolved to the newly established Electricity Regulatory Board (ERB) which became functional in 1998.

[9] From 1997, KPLC began to focus exclusively on transmission and distribution, while a separate entity known as KenGen (formerly Kenya Power Company—KPC) took over all public power generation activities.

[10] The Energy Act of 2006 provided that the ERC would be funded through not only electricity levies but also fuel levies from downstream petroleum regulation, enhancing its financial independence.

- partially privatize KenGen through an initial public offering of 30% of its equity through the Nairobi Stock Exchange.

Unlike the first wave of the reforms, which were mainly donor-driven, the second set of reforms was initiated and led locally. By 2007, most of these measures were implemented,[11] including KenGen's listing on the Nairobi Securities Exchange in 2006 (Eberhard, Gratwick, and Kariuki 2018; Kapika and Eberhard 2013). In 2008, the Kenya Electricity Transmission Company Limited (KETRACO[12]) was established to focus on the construction of new transmission projects, facilitated by funding from the government and donors through concessionary financing, while KPLC retained responsibility for operating the grid. The GDC was also established but has been less than successful in its mandate to lower the risks of geothermal development for the private sector investment (Godinho and Eberhard 2019).

In the same year, Kenya's Vision 2030 (encompassing social[13] and economic goals) set a new generation target of 23,000 MW by 2030 (up from 1,310 MW in 2008) (Eberhard, Gratwick, and Kariuki 2018). In 2010, the government began work on a nuclear power project through the Kenya Nuclear Electricity Board (KNEB), recently re-established as Nuclear Power and Energy Agency (NPEA) in 2019, an autonomous institution within the MoEP (Government of Kenya 2019). The initial aim was to generate 1000 MW of nuclear energy by 2023, but by 2019 little progress had been made. However, these changes increased investor confidence and interest in Kenya's energy sector.

In September 2013, the Ministry of Energy and Petroleum (MoEP) launched the "5000+ MW" program with the goal of bringing at least 5000 MW of power online within forty months, and IPPs were prioritized to play a central role (Government of Kenya 2013). The program was heralded by the Kenyan government as the means to "transform Kenya," by providing adequate generation capacity at a competitive rate (MoEP 2013). However, this program has proven difficult to implement and points to some weaknesses in coordination and planning in Kenya's power sector. In 2017, the government announced plans to halt the program, citing concerns about depressed demand and expensive loans (Kenyan Wall Street 2017). Meanwhile, at the generation level, the ERC in 2014 affirmed that "electricity generation in Kenya is liberalized," with IPPs given an opportunity to enter the sector and compete alongside the state-run KenGen (Eberhard, Gratwick, and Kariuki 2018).

Nonetheless, a competitive market structure has been set in motion with the passing of the Energy Act 2019, providing the legal basis and institutional frameworks to facilitate a competitive wholesale market structure in the country.

Private Sector Participation

Private sector participation in generation has been practiced for over two decades in Kenya and its share of installed generation capacity rose to 36% by 2020, contributing about 26% of production. With the increased IPP commitment and shift to renewables, IPPs are expected to play a significant role in the future.[14]

[11] Exceptions were the development of a fully competitive market structure and the ambitious rural electrification target.

[12] KETRACO currently owns over 42% of the total transmission network and is poised to become the dominant transmission company once the committed transmission projects are completed.

[13] Rural electrification efforts aimed to bring electricity to every home in Kenya with interim targets set for 2013 and 2022 (these have since been shifted).

[14] Of the near-term capacity envisioned in the 5,000+MW program, the majority (70 percent) is expected to come from the private sector with KenGen and GDC developing the balance.

Progress in Setting up IPPs

The first wave of privately financed generation dates back to 1996 involving the procurement of two diesel IPPs: Westmont (46 MW) was sponsored by a Malaysian firm, and Iberafrica[15] (44 MW) (Eberhard et al. 2016). With a tenure of seven years—longer than that of most emergency power producers (EPPs)—these first two IPPs were considered stopgap measures for addressing drought-induced hydroelectric shortages and the delayed construction of projects envisioned in the least cost power development plans (LCPDP).

The second wave of IPPs—during 1997 to 1999—occurred amid a move to reform and liberalize the sector. In 1996, KPLC ran international competitive bids (ICBs) for two projects—Olkaria III and Kipevu II—which came to be known as OrPower4 (with varying MW/geothermal) and Tsavo (74 MW/diesel) respectively. Although both projects were procured via ICBs, it is noteworthy that only three bids were received for the Tsavo plant and two for what would become OrPower4 (Eberhard et al., 2016). KPLC subsequently also ran ICBs for the thermal IPPs—Rabai, Thika, Gulf, and Triumph—obtaining progressively more competitive prices. Drought conditions and worsening hydrological conditions in the decade 2000–2010 led the MoEP to directly engage three international EPPs (Aggreko,[16] Cummins, and Deutz) for a combined 105 MW rental capacity during 2000–2001. In 2012, there were 120 MW of EPPs: to date, all these have since been decommissioned. Recently, several projects have been procured through a renewable energy feed-in tariff scheme. By end of 2020, Kenya had 17 IPPs with an installed capacity of 1,013 MW, thus increasing the security of supply.

Private Management Contract—Manitoba Hydropower

Reforms also led to Kenya's first private management contract in the power sector from 2006 to 2008. Following a review of tariffs and performance of KPLC that revealed inefficiencies, KPLC was put under a private management contract—which was seen as a means of severing political and bureaucratic influence over utility staff by government officials (Godinho and Eberhard 2019) and was also a condition to unblock US$152 million from the World Bank (Africa Business Insight 2016).

A two-year contract was awarded to Manitoba Hydro in May 2006 after a competitive tender. The contractor provided a three-member senior management team[17] comprised of the CEO, finance, administration and technical experts, and all other departments were re-aligned and merged to report to the three functional heads. The contract continued the positive trends of reducing system losses and improving collection rates, with the latter reaching close to 100%, while losses decreased further to 16.6% in 2008.

At the same time, profits increased, and a healthy debt-service coverage ratio was maintained. The utility increased new customers by 258,134 in two years from the previous 40,000 per annum. The contractors also introduced preventive "live-lines maintenance" without switching off power, which greatly improved reliability. In 2008, the government decided not to renew the contract citing the realization of the set objectives for contractors. Despite this success, some stakeholders interviewed, however, pointed to the "tension and

[15] Iberafrica renewed its contract in 2004 and increased its capacity progressively to reach 108 MW in 2015.

[16] In 2006 Aggreko would be called upon again to provide 80 MW and further increases to 290 MW by 2009.

[17] Worth noting is that the new management did not terminate any of the existing staff but focused on training and coaching local staff to improve work culture.

conflicts that arose in 2007 (at the onset of a drought) between the MoEP and contractors, regarding direct procurement of IPPs and EPPs desired by the government, which the contractors rejected," for the non-renewal of the contract. Others cited "the anomaly of government continuing to provide subsidies to KPLC while the contractors were promising financial viability, and the fear of possible staff retrenchments." Though the management team was not well-liked by local KPLC staff, the two-year period allowed them to build internal capacity and a greater degree of independence from political influence, as well as a sense of purpose and ownership over the new trajectory of power sector reform and development in Kenya. As part of its exit strategy, Manitoba Hydro partnered with KPLC to form Kenya Power International—an offshoot of KPLC that provides consulting services across Africa and the Middle East.

Governance Framework and Structure in Kenya Power (KPLC)

KPLC's governance is framed by key legislation and policies, including the Companies Act 2015 Cap. 486, the Capital Markets Act 2012 Cap. 485A, the State Corporation Act 2012 Cap. 446, the Code of Governance for State Corporations (Mwongozo), and the Board Charter, all of which outline the structure, composition, roles, responsibilities, and functions of the company, the board, and committees. KPLC is a listed company with 49.9% of its shares[18] publicly traded on the Nairobi Securities Exchange and the remaining 50.1% owned by the government of Kenya. This capital structure makes it a mixed capital enterprise (MCE), thereby creating new principal-agency relationships.[19]

The corporatization reform and streamlining of board composition to include independent directors is meant to check excesses of government-sanctioned political influence as well as to protect the interests of private shareholders (principals), without which the company risks permeation of rent-seeking moral hazards.

In addition, the board of directors (BOD), as an internal control mechanism for agency problems helps to monitor and evaluate the performance of top management. It helps shareholders to control possible information asymmetry, management perquisite consumption,[20] and entrenchment[21]—traditional agency problems that affect performance.

However, since private shareholding is thinly and widely spread, with no large-block stockholders, in practice "there has been minimal-to-no representation of private sector interests on the board" as intimated in one of the interviews. "The government as majority shareholder has a firm grip on the board." The lesson from this governance failure is that continued government influence adversely affects post-privatization company performance.

Mwongozo (The Code of Governance for State-Owned Corporations)

The Mwongozo[22] (Government of Kenya 2015) was born out of the realization that state-owned enterprises (SOEs) had, over the years, and with new challenges, not

[18] Most of these are owned by institutional investors (banks) and local individual investors.

[19] For this reason, the company is also obligated to comply with the Code of Corporate Governance Practices for Issuers of Securities to the Public 2015, and the 2002 Capital Markets (Securities, Public Offers, Listing and Disclosure) Regulations.

[20] Short-term cost-augmenting activities by managers designed to enhance non-salary income or provide other on-the-job consumption.

[21] Refers to actions of management that reduce the effectiveness of control mechanisms designed to regulate management behaviors

[22] Swahili for Guide or Regulations was developed in 2014 and is anchored in the Constitution of Kenya, 2010 Articles 10, 73 and 232 that espouse national values and principles of good governance –integrity, efficiency, and effectiveness, and carries the same legislative importance.

operated at expected levels owing to weak governance structures and poor leadership in the management of public resources. The code has been adopted as a state-motivated reform to create a transformational mindset in the way business is conducted to increase efficiency and accountability in the use and deployment of scarce resources. One of the governance challenges faced by SOEs is insufficient competence in boards of SOEs, as well as embedded political interests, arising from an opaque appointment process. A professional and independent board is more likely to safeguard an SOE from political interference, lead to more efficient operations through a well-defined strategy, and could ultimately result in increased value-for-money to the shareholders. The Code of Governance,[23] in relation to Kenya and KPLC, provides a framework for improving corporate governance by allowing for the appointment of professional boards with well-defined skill sets; the undertaking of board inductions and evaluations, and requires regular performance reports.

Drawing from its past aid embargo experience and the desire to exploit energy resources to spur growth in the economy, the government implemented the Mwongozo to streamline governance and attract investments. Despite the previous caveats around ongoing political interference in KPLC's board, the Mwongozo code is a positive reform initiative to inculcate good governance principles in all SOEs. It promotes greater autonomy, transparency, accountability, and leadership focused on performance, for the purpose of improving service delivery unlike in other countries in the region. This initiative has also created a better investment climate and gives greater regulatory certainty, which has catalyzed the growth of IPPs as well as greater certainty of shareholder investments in KPLC.

Accountability, Transparency, Disclosure, Risk Management, and Internal Control

While KPLC maintained a strong financial performance for many years, in recent times its solvency has declined and, as such, the Auditor General noted a breach of commercial borrowing covenants in 2018 by KSh59.9 billion.[24] Although lenders waived their rights to demand payment owing to breach of debt covenants, it signals increasing financial stress.[25] This breach also means that the company violated the capital markets[26] listing obligations in regard to maximum debt capacity and solvency[27] (KPLC 2018, 112–17).

In July 2020, the board of KPLC was forced to resign for failing to disclose to the government the magnitude of financial distress resulting from an inadequate tariff review, excessive debt levels and high operating costs associated with the access and network expansion program, overcommitment in new IPPs (the 5,000+MW program), and financial mismanagement in operations, threatening the financial sustainability of the company (Business Daily 2020a; Muli 2020). In sum, the intervention of the auditor-general and penalties by the CMA underscore the strength of the independent governance system in Kenya and in KPLC, occasioned by corporatization and privatization reforms. It can also be

[23] The Code has also provided for reform and reduction in the size of the boards and the increase in the number of independent board members to reduce conflicts of interest, which is a game-changer in boardroom affairs. It further provides a platform for addressing shareholder rights and obligations with the aim of ensuring that sustainable performance excellence becomes the hallmark of SOEs.

[24] KPLC's total borrowing as at end of 2018 amounted to KSh113 billion.

[25] In July 2018, majority of KPLC senior management were arrested and charged at the High Court with alleged corruption relating to procurement violations and material misstatement of financials (information asymmetry) and subsequently terminated.

[26] Although the Capital Markets Authority (CMA) imposed penalties as confirmed from interviews, these were not publicly disclosed.

[27] At the end of 2018, the company's current assets of KSh54.6 billion were less than current liabilities of KSh106.3 billion, resulting in a negative working capital of KSh51.6 billion

argued that the intervention of the Auditor General underscores the failure of the incentive-compatible regime which requires further strengthening.

The Capital Markets Governance, Listing Rules and Regulations

KPLC's listing on the main investment market alters governance arrangements by introducing new and wider stakeholders—such as customers, the public, the media, investors and other market participants, unlike in traditional state-owned enterprise (SOE) governance structures. A strong discipline of private capital is introduced through stock listings and bonds by increasing the levels of scrutiny on governance and management processes. One such discipline is for listed companies appointing a person who is a certified director from an institution recognized by the CMA. This standard has the potential to eliminate would-be incompetent political appointees in SOEs and reduces conflict of interest by government, since qualification requirements are pre-defined by an independent institution. In addition, the capital markets regulations introduce an element of penalty, liability and sanctions to the company, or any company director, or management staff (jointly or severally) who contravenes the set regulations—*legal effect.*[28]

The CMA also provides for potential suspension and delisting of the company. This incentive mechanism compels directors and management to strive for improved performance. Related to this are requirements to improve financial performance, pursuant to the listing and disclosures regulations (CMA-Kenya 2002)[29]—the *performance effect*. It also sets as a mandatory requirement for the company not to be in breach of any of its loan covenants, particularly regarding the maximum debt capacity.

The *accounting effect* is that the existence of strong accounting standards does improve the investment performance and climate of a company as far as shareholders are concerned. Therefore, the CMA and financiers require that an external audit is done on the financial covenants every year in accordance with international financial reporting standards (IFRS) and before publication of results in the main media outlets.

Electricity / Energy Regulation and Regulatory Regime

Since its establishment as an electricity regulator in 1998, and then as an energy regulator in 2007, the Energy and Petroleum Regulatory Authority (EPRA)[30] as a principal has played a crucial role in balancing decision-making in the sector, especially between privately owned IPPs and the publicly owned generation company KenGen. The EPRA has licensed new capacity, thereby assisting to create security of supply and approving near-cost-reflective tariffs for KPLC. However, recent involvement by the government in sector regulation and planning is undermining regulatory independence and effectiveness in regulating market entry, tariff reviews, and system planning.

Following an earlier pronouncement by President Uhuru in 2016 to reduce the cost of power (Ngetich 2016), a task force team was formed, headed by the Ministry of

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[28] Since funding is usually sourced from local and international financiers, the legal burden of default compels management to minimize wastage and not engage in risky investments.

[29] Specifically, the First Schedule of the Capital Markets (Securities) (Public Offers, Listing and Disclosure) Regulations, 2002 (Amended 2016) sets out minimum requirements for listing and continuing obligations of the listed entity to have a clear future dividend policy and a track record of profitability and future prospects—including adequacy of working capital.

[30] In 2019, the EPRA was reconstituted as a successor to the Energy Regulatory Commission (ERC). However, its mandate has been widened to include upstream petroleum and coal regulation.

Energy, to review all existing PPAs, to establish revenue requirements for each, to explore possibilities for mutual termination (or compensation payments for termination), and to compare PPA costs with similar plants regionally and globally (Juma 2021). The resulting major recommendations among ten others are: (i) that existing PPAs should be renegotiated, (ii) pipeline/unconcluded PPAs should be canceled, (iii) the cost of electricity be reduced by 33%, and (iv) restructuring KPLC into a commercial entity that is both profitable and providing cost-effective electricity (Government of Kenya 2021). While the review team appears to have focused on promoting affordability and equity objectives for consumers rather than overall sector sustainability, there is no clear pathway for how the regulator will implement these recommendations without ensuing legal suits. Even if the recommendations were implemented as is, they would plunge the sector into a financial crisis and therefore need a rethink.

In addition, in a bid to implement the government's recent policies geared toward industrialization and employment, the regulator has been directed to heed policy and not to increase tariffs. The government's emphatic argument as intimated in interviews has been that "the law/regulation follows policy and hence any inconsistencies in it are an area for law reform and not the reverse." Specifically, the EPRA has been pressured to set low tariffs that reflect policy objectives rather than economically efficient prices. Regardless of political influence, at the time of data collection, the regulator was also accused of an inability to stand firm in exercising its legal mandate to determine cost-reflective tariffs, owing to fear of job termination signaling weakness in institutional and leadership[31] capacity. The government has seized this weakness to push for a move away from cost-reflective tariffs. This signals a potential decline in the role of the regulator and the energy sector.

Analysis and Findings: Operational Performance of KPLC

An empirical and analytical analysis of the operational performance of KPLC, considering five key performance measures which mirror the enduring challenges facing utilities: access to electricity; adequate and reliable supply; efficiency; affordability, and financial sustainability is presented below.

Access and Electrification

Kenya is implementing an aggressive electrification program that has more than doubled the number of grid connections in a span of just five years with the aim of achieving universal access to electricity by 2022. KPLC has connected, on average, about a million new customers annually during the last four years (2014–2018) and had about 6.7 million customers as of June 2018.[32] About 65% of these new connections are poor households that consume less than 15 kWh monthly (Fobi et al. 2018; Lee et al. 2016). The government has developed programs of subsidizing connection costs through budget transfers[33] of about US$700 million.

As a result, national electrification rates are currently 75% (84% of urban and 71% rural) (World Bank 2018). However, in a bid to meet the last mile connectivity project

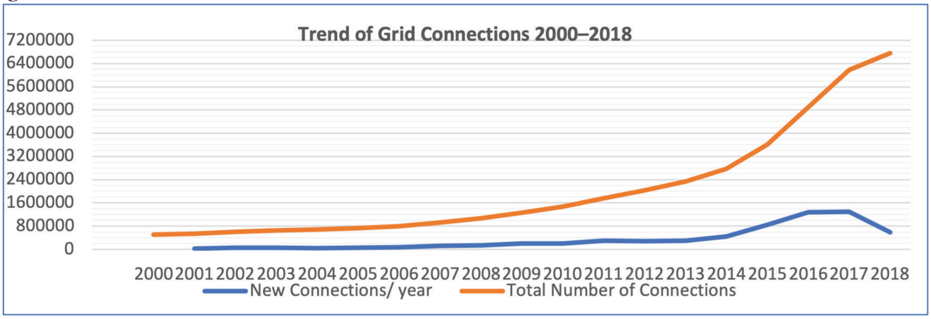
[31] A new Director General and Board was appointed in January 2022 who appear to be performance driven.

[32] This performance exceeds what South Africa achieved at the height of its electrification program—less than 500,000 connections per annum—but still one of Africa's most successful efforts, raising access rates from about 50% in 1994 to close to 90% today.

[33] Under the LMCP, households pay a reduced connection cost of KSh15,000 (US$150), and the Government subsidizes the rest. An innovative output-based aid mechanism (OBA) was adopted for slum electrification.

(LMCP) targets, KPLC has also had to divert resources to access-related infrastructure, and these amounts swelled over time as the utility tried to connect people living farther away from the grid. Billings, too, (consumption and net revenue per customer) and loss reduction have deteriorated. Whereas the Last Mile[34] program obliged the government to reimburse KPLC for maintenance and connection costs, the transfers have been insufficient creating a huge viability gap for the utility.

Figure 3. Number of Connections to the Grid



There is a broad lesson here. Progress in electrification requires national goals and targets, planning, and subsidies for connections. However, as more and more low-income households are connected, especially in deep rural areas, the issue of affordability arises (which is examined later). Likewise, consumption remains low and is generally insufficient to either pay for grid strengthening and extensions, or the operating and maintenance costs of these services. It shall be shown how this has impacted KPLC's financial performance. For high levels of electrification to be sustainable, some level of cross-subsidies is generally required for low-income households.

Adequate and Reliable Electricity Supply

Over the past two decades, Kenya has been one of the most successful countries, a leader in Africa, in taking bold steps towards enhancing its security of supply. Particularly, since 1990, net generation capacity has more than tripled to reach a current level of just over 2.8 gigawatts (GW). Kenya has also diversified its power system from reliance on almost exclusively hydroelectricity to a more balanced mix of hydropower, geothermal and oil-fueled generation, and, in recent years, also wind and solar energy. Kenya's previous reliance on hydropower led to repetitive supply crises in the 1980s and 1990s. With the aid embargo, and unable to access funding for new generation capacity, the country faced severe supply constraints resulting in the contracting of expensive emergency thermal power, which, in turn, increased the cost of supply as it waited for the commissioning of the state-owned and IPP plants. The country revised its approach to generation planning and adopted reforms to attract private-sector investments into the sector. Following the unbundling of generation from transmission and distribution, a new and better-coordinated planning and procurement system ushered in a series of IPPs worth over US$3.0 billion that helped to increase capacity

[34] It is worth noting that the early structural reforms of the sector did not materially impact the pace of electrification and it was necessary for the government to adopt additional and more ambitious targets for universal access, backed by dedicated financing and donor support, without which access rates would have remained low. In addition, regulatory reforms in the early 2000s have created a strong enabling framework for growth of off-grid electrification efforts in Kenya as a leader in access in the region.

and translated into improved reliability of supply. While generation capacity appears to secure supply for the interim future, not enough investments have been made in the transmission and distribution infrastructure to unlock demand. There is also a perception that private investments have led to high tariffs, an issue that is yet to be addressed.

Figure 4. Trend of Growth in Installed Generation Capacity (MW)

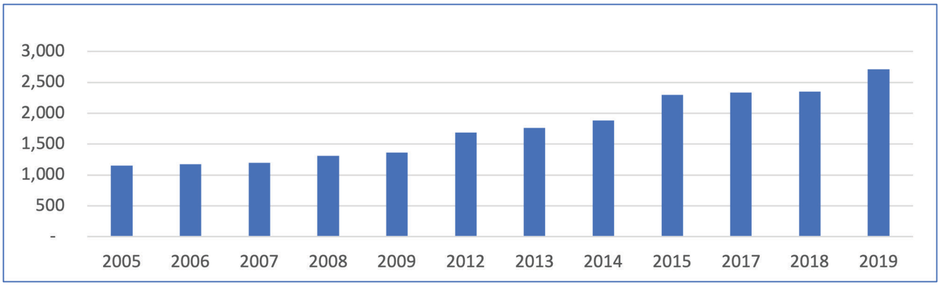
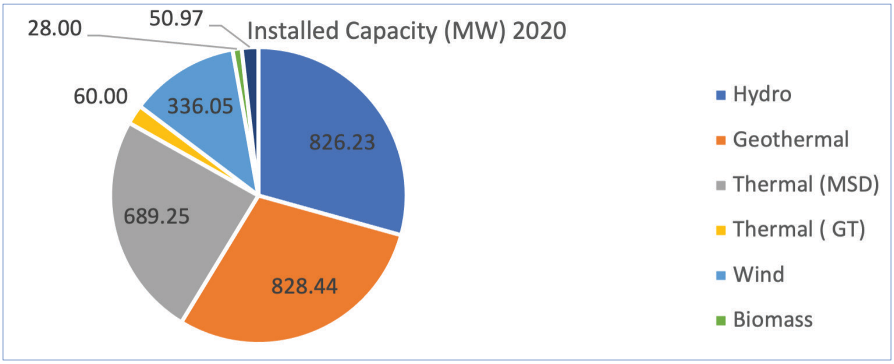


Figure 5. Installed Generation Capacity by Energy Source

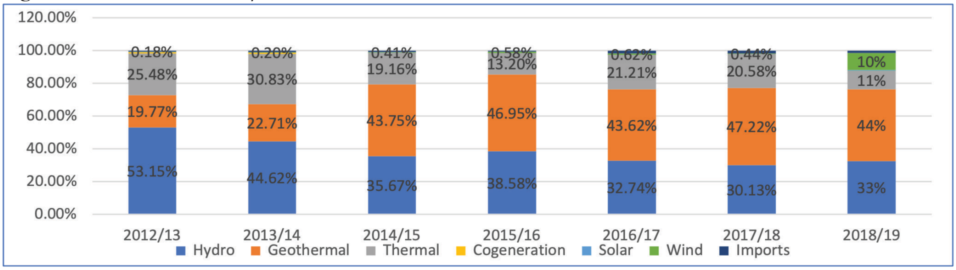


Generation Mix

Kenya's security of supply has improved through marked diversification of the energy mix. With restructuring, the country sought to de-risk geothermal resources managed by a corporatized GDC. Currently, geothermal[35] capacity is over 828 MW contributing more than 44% of electricity production, and this share has been growing since 2014, helping to displace the more expensive thermal generators. Overall, there is a shift towards more renewables whose share is now amongst the highest in the world.

[35] Following the commissioning in 2015 of the latest large geothermal plant of 280 MW at Olkaria, consumers saw a reduction in fuel costs. The share of hydro has declined from above 50% in 2012 to about 33% today. The recent (2019) commissioning of the Lake Turkana Wind Project-310MW, has increased the contribution of wind to 10% and was projected to increase to about 12% beyond 2020.

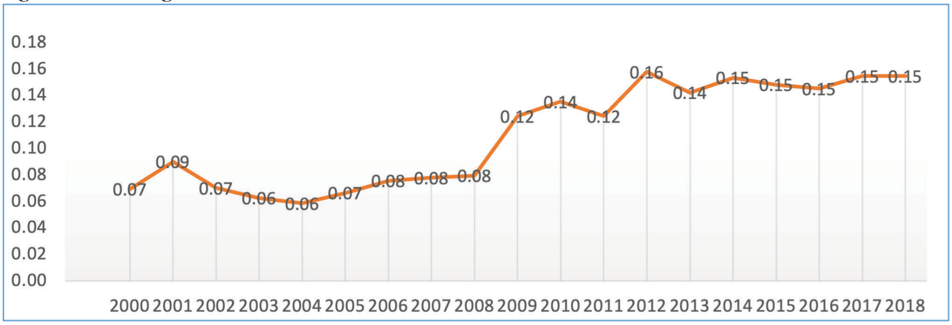
Figure 6. Trend of Kenya's Generation Mix (MWh)



Affordability

Despite the relatively high cost of electricity in Kenya, the cost of a subsistence volume of 30 kilowatt-hours per month is well under 5% of the budget of the poorest 40% of households (ESMAP 2018). Electricity appears to be affordable with lifeline tariffs (KSh10 for 100 kWh) and subsidized connection charges. However, electricity consumption has not risen as sharply as the number of connections.

Figure 7. Average Tariff (¢/kWh) Trend



Source: Author's creation based on KPLC and EPRA data.

RoR is applied in determining base and retail tariffs to guarantee the financial integrity of KPLC and the sector,[36] profits are monitored and there is no incentive to reduce service quality. This allows fixed costs to be allocated to customers across a stepped tariff structure and recovered through energy sales. This has stabilized the sector's financial standing although recent disruptions/political externalities are threatening this. While the regulator is meant to conduct a tariff review every three years, only four reviews have been carried out—1999, 2008, 2013, and 2018. In 2018, a single-year review was granted rather than the prescribed multi-year tariff and no further review has been conducted since.

Although reforms led to the establishment of an independent regulator, regulatory independence in tariff determination is increasingly eroded as the government influences tariff decision-making to achieve political goals. Often under the pressure of elections, the government has issued Presidential and Ministerial directives halting the automatic tariff adjustment and review mechanisms. A top manager in an interview decried this as: "the regulator is heavily influenced by the government—doesn't provide adequate revenue requirements which threatens the commercial viability of KPLC." Another respondent had

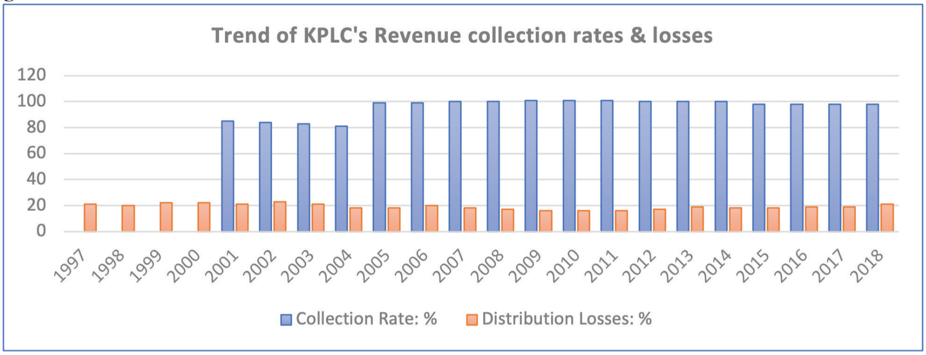
[36] KPLC is the designated offtaker for IPPs, and ensures stable financial flows to the private sector

this to say "ERC/EPRA as a principal is facilitating the financial collapse of KPLC." Since the ERC/EPRA Board of Directors and Director General are appointed by the President and Minister of Energy and Petroleum, they are "guided" not to implement certain tariff increase decisions as a trade-off for keeping electricity prices low (social objectives/goal) to gain electoral votes and if such "guidance/directives" are not adhered to, may lose their jobs. Such directives, however, negatively affect the financial viability of the utility and the entire energy sector. Hence, in a weakened institutional governance setting, governments can influence pricing decisions even under partial post-privatization conditions.

Efficiency

For this article, efficiency is assessed in a limited way using two KPIs—system losses and revenue collection rates. Following the resolution of the tariff imbalance between KPLC and KenGen, the KPLC was put on a performance contract (PC)[37] in 2004 and KenGen in 2005. The introduction of a private management contract in 2006 (Manitoba Hydro) helped to push collection rates to 100% and technical losses declined to 17% by end of the contract in 2008. The recent connectivity drive for universal access, however, has dented this stellar performance as collection rates have declined to 98% and losses have risen to highs of 21% largely due to non-payment of bills and power theft. IT innovations in prepayment and smart advanced metering infrastructure (AMI) or automatic meter reading (AMR) metering, coupled with the breakthrough in mobile telephony fintech[38] and pay-as-you-go payment options have helped to keep revenue collections close to 100%. However, government entity arrears remain a sticking challenge. While reforms like the management contract enabled greater efficiency improvements, these benefits are being eroded by governance weaknesses associated with political influence. It remains to be seen whether KPLC will recover from this recent decline.

Figure 8. Trend of Revenue Collections and Distribution Losses



[37] The PC, as a new governance-monitoring tool, had an immediate impact on collection rates, which increased to almost 99% in 2005 from 81%, and transmission and distribution losses reduced to 18%.

[38] Telecoms have devised simplified applications that enable customer to pay bills anytime and anywhere creating greater convenience in bill payment.

Financial Sustainability

KPLC's Financial Performance

Gross revenues grew from KSh 23 trillion in 2004 to KSh126 trillion by 2018, representing an average growth rate of 10% per annum over the fourteen-year period. However, with the implementation of the Last Mile Connectivity Program and the connection of more low-demand customers to the grid, the average consumption per customer reduced by an average of 15% per annum over the period. As a result, net consumer revenues from both industrial (anchors) and domestic customers declined owing to the rise in cheaper captive and reliable renewable sources, such as solar photovoltaic (PV), and power thefts respectively (Murefu 2020). There is, however, growing concern that the growth of other distributed generation (DG) systems will shrink the market share of the traditional Tesla-Westinghouse "bulk-grid" model (BGE) which forms the bulk of Kenya's power market structure. However, the growth of DG creates massive opportunities for more reliable, cheaper, cleaner and closer sources of renewable energy especially because of the wind and solar resources in various parts of Kenya, allowing the country to leapfrog expensive fossil fuel technologies and pursue a climate-friendly, needs-oriented power strategy for low-carbon growth while protecting livelihoods and the environment. Nevertheless, regulatory reform that enabled tariff increases has contributed to revenue growth, with the most impactful tariff increases occurring in 2008 by 24%, as well as in 2011 and 2013.

Figure 9. Revenue Growth (KSh millions)

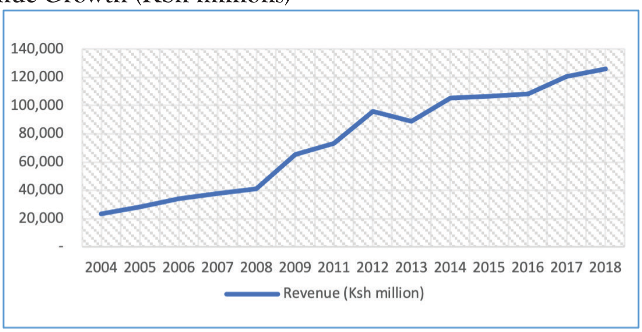


Figure 10. Gross and Net Profit Margins 2004–2008

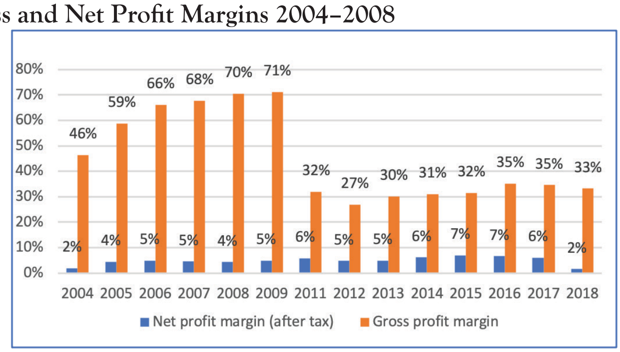


Figure 11. KPLC Debt-Service Coverage Ratios

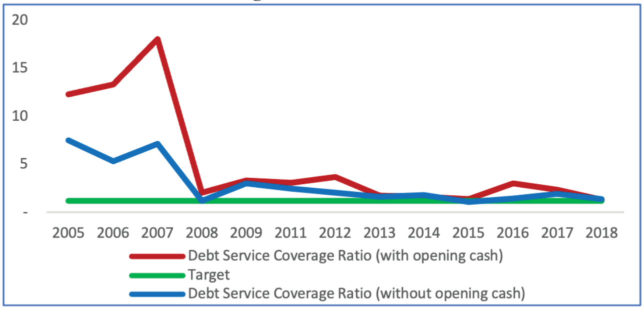


Figure 12. Debt to Equity Ratio

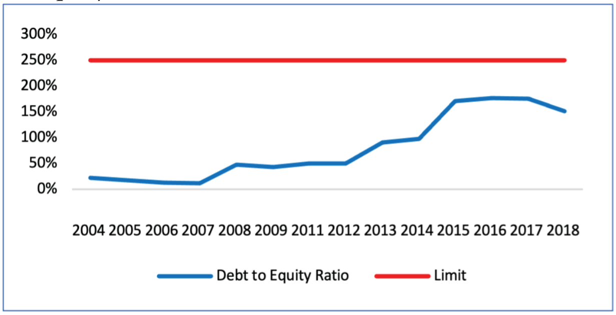


Figure 13. Cash (Self-Financing) Ratio

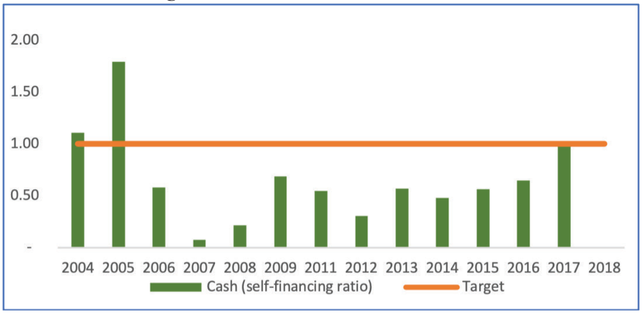


Figure 14. Current Ratio

Figure 15. Interest Coverage Ratio

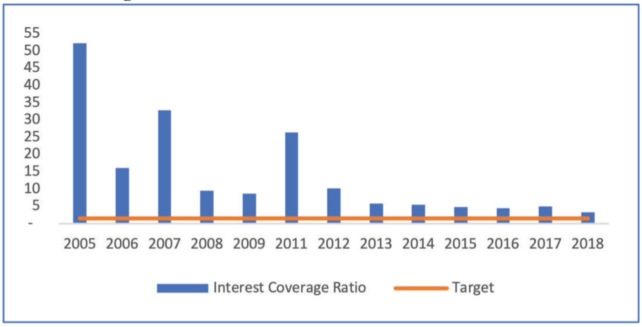
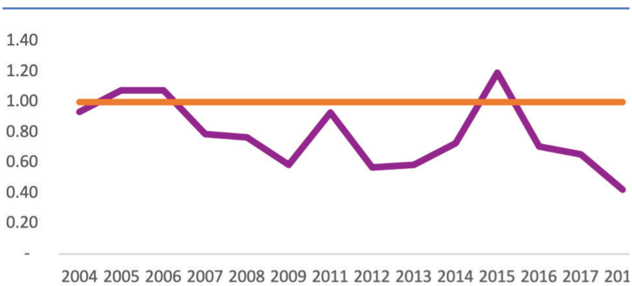


Figure 16. Acid Test (Liquidity Ratio)



However, the net profit margin declined to 2% in 2018 owing to a delayed review of retail electricity tariffs,[39] increased operating costs associated with the access programs, delayed fuel cost recoveries, increasing financing costs and doubtful electricity receivables, which negatively impacted cashflows. In nominal terms, profits reduced by 92% from US$49 million in 2018 to US$3 million in 2019 (KPLC 2018).

As a result, recently, KPLC has issued three profit warnings (cautionary earnings statements 2018, 2019, and 2020) to its shareholders, citing depressed economic activity and demand, inadequate tariff reviews, and increasing financing costs (Anywanza 2020). In order to reverse and stabilize its financial position, the company has sought external debt relief in the form of a twelve-month moratorium on its foreign currency-denominated loans and is in talks with local banks to reschedule its multibillion-shilling debt (KSh68 billion), which has seen its profitability dip to the lowest in sixteen years (Business Daily 2020b).

The solvency ratios—debt-service coverage ratio, debt-to-equity ratio, and interest coverage ratio—have been positive but there is a declining trend. The company obtained additional facilities[40] in 2015 which increased its debt by 87% year on year. This led to a 39% drop in the DSCR to 1.08, which was marginally below the target of 1.2. In the same year, the company breached its loan covenants and was vulnerable. Further, recent declines in cash flows have made it susceptible to default, unable to service its debt, and seeking

[39] The 2013 schedule of tariffs was for a 3-year period with a new tariff expected from June 2016, but this did not happen. Considering that the average revenue requirement, which drives the tariff computation, takes into account the total transmission and distribution assets, failure to review and issue a cost reflective tariff meant that the revenue requirements for 2017–2019 was based on the size of the grid assets as at 2013 and did not take into account the significant expansion of the network in the preceding period, thereby creating a huge financial viability gap for KPLC.

[40] KPLC acquired a long-term syndicated loan of KSh50 billion from The World Bank, AFD, Standard Chartered Bank and Rand Merchant Bank

waivers from its lenders (Business Daily 2020b). To date, KPLC's total debt is estimated at US$1 billion.

An analysis of the liquidity ratios—cash (self-financing ratio), current ratio, and acid-test ratio—shows a declining trend in performance, the increasing financial risk of the company, and is unsustainable.

This analysis demonstrates that while *partially* privatized utilities, such as KPLC, can perform better than state-owned utilities if they still have a majority government shareholding, financial governance systems may be vulnerable to being undermined by governments.

Conclusion and Policy Implications

At a policy level, Sessional Paper No. 4 of 2004, laid a firm foundation for legal and institutional reform of the power sector in Kenya. The policy and new legislation facilitated the restructuring of the power market and unbundling of KenGen from the vertically integrated KPLC to: enable a competitive power market for generation, distribution, and supply of electricity; the introduction of a lifeline tariff; the partial privatization of KenGen through an IPO offering 30% stock to the public; and the creation of other corporatized institutions (namely, KETRACO, GDC, and REA/REREC) as agents of government that have since also impacted the performance of KPLC in various ways. Despite the above structural reforms, further reforms are needed to separate the Transmission System Operator from the distribution segment allowing for more horizontal competition, liberalization of the retail segment, and greater cost and technical efficiencies in the sector.

For a long time, Kenya has had a robust planning system buttressed by optimal LCPDPs designed by KPLC expertise, and well linked to procurement of new generation capacity. However, increasing government influence in planning activities in recent years has resulted in the proliferation of directly negotiated power procurements that are poorly aligned with LCPDP leading to excess supply and costly power for consumers. This trend should be reversed by enacting legislation and stronger regulations that revert the technical planning mandate to the competent KPLC team instead of the line Ministry. Despite the Kenya government's efforts to revise PPA prices downwards, this has only served to dampen investor confidence in the energy sector, and alas, the recommendations for lowering tariffs by 33 percent by end of 2021 have proved difficult to implement. The key lesson here is that in a partially privatized utility, where the power off-taker is government-owned, planning choices are likely to be unduly influenced by political considerations and these can undermine utility performance. This is mainly because the state as a major shareholder and its policy actors have rent-seeking objectives and see energy as an easy source of revenue through non-transparent directly negotiated deals. Since the then regime had set out an ambitious generation expansion plan to build new power plants via the 5000+MW program, government shareholding could create discretionary pathways for corrupt behaviors to seep through as they source new power developers, a practice that previously was not possible under KPLC planning processes. If equity objectives are to be realized, then government policy should provide for direct subsidies to lower tariffs in the interim.

An examination of the different aspects of performance commencing with the imperative of access, reveals that the government of Kenya (GoK) has driven the access and electrification development goal, through an aggressive campaign, to levels higher than in any other country in the region. However, this ambitious program is now threatening the financial sustainability of KPLC. Consequently, the principal's continued interference has adversely affected KPLC's liquidity ratios to the extent that the utility has breached its debt covenants. Thus, again, despite KPLC being partially privatized, its financial governance controls have been undermined, enabled by the government's ongoing majority shareholding. Government should instead fully reimburse KPLC for electrification costs and strengthen the principal-agent obligations and relationship by minimizing influence in KPLC's governance systems.

How have the reforms impacted Kenya delivering an adequate and reliable power supply and altered governance arrangements? The unbundling of KenGen from KPLC created an autonomous agent (KenGen)—with delegated authority to procure and contract new publicly funded generation capacity as well as in opening the market for more IPPs. The rebalancing of tariffs in the early 2000s helped to create bankable balance sheets for KenGen and KPLC, allowing them to attract private investment in generation on commercial terms and ensuring the security of supply. The unbundling has also removed potential conflicts of interest between KPLC, KenGen, and IPPs. However, since KPLC is largely state-owned, there is continued proliferation of expensive unsolicited power projects and therefore structural reforms seem inadequate and need to be complemented by additional governance measures like stronger public entity management legislation. There is also a need to change the political economy of the utility to improve transparency and information reporting.

Regarding tariff affordability and cost reflectivity, regulatory reforms established the regulator and have since 2007 played a crucial role in balancing decision-making in the sector, especially between IPPs and the publicly owned KenGen. This has helped to improve cost reflectivity especially in licensing new capacity. Unlike in previous periods when regulatory independence in tariff-setting was respected by all stakeholders, recent involvements by the government/principal in issuing directives to halt or postpone tariff reviews, have caused a decline in the financial performance of KPLC and have undermined regulatory independence. Regulatory independence should be enhanced by enacting stronger legislation that deters direct political directives. This can be done by developing and following a clearly pre-determined legal process for reversing regulatory decisions known to both entities rather than ad hoc termination of directors and directives. This would increase the independence, certainty and credibility of regulatory decision-making for consumers, investors and utilities.

Until 2018, KPLC maintained a healthy financial performance record enhanced by the securities exchange listing obligations and tariff increases. However, in recent years, its financial state has declined significantly. The reversal of the 2017 tariff review had a direct, negative impact on the company's revenues. The company fundamentals have fluctuated with significant declines to below minimum thresholds, indicating the financial vulnerability of the company occasioned by the government's and the regulator's failure to implement cost-reflective tariffs. Additional governance measures like transparent transfers of social programs and regulatory predictability should be enforced to protect the utility's financial standing.

From a capital structure perspective, the listing of KPLC on the NSE has established contractual relationships not only with the board and shareholders but has also allowed the entry of other principals interested in improved performance. While the listing as an external governance mechanism incentivizes[41] management and KPLC to remain profitable, solvent, and with adequate working capital, a confluence of factors—such as non-cost reflective tariffs;

[41] Since the CEO and some directors (agents) own equity stock in the company, their interests are aligned with those of the owner (government), helping to drive performance.

overborrowing to finance the aggressive universal access programs; the expensive cost of capital; delayed government subsidies, and internal mismanagement—have led to a decline in its financial performance. In this way, the principals' enduring conflict of interest in the governance of KPLC has had detrimental financial repercussions not only on KPLC but also on the capital markets. This requires a review and rethinking of the efficacy of the current performance contract, improving the codes of governance to increase the level of scrutiny, and reporting as well as enhancing the public entity management legislation to provide for more stringent monitoring controls.

In conclusion, the Kenyan case provides fascinating insights into how legislative, regulatory, structural, and governance reforms can create incentives for improved utility performance. However, the role of the principal, including in a mixed capital enterprise such as KPLC, remains critical. When the relationship between principal and agent is well understood and the agent is properly incentivized, performance improvements are possible. When the principal undermines or muddies those incentives through conflicting political interventions, performance improvements can be reversed.

I acknowledge that there are multiple principals and agents creating a web of relationships. Some of these relationships manifest in the form of lenders to government or consumer groups or private sector innovation technology hubs offering various services to the utility, government and regulator. These play various important roles and need to be studied further depending on the focus of the study.

Data Availability Statement

The data presented in this study are available on request from the corresponding author due to ethical considerations. The qualitative data is kept private as it would reveal the identities of interviewees. Quantitative data can be obtained from reports cited in the references.

Disclosure Statement

In accordance with Taylor & Francis policy and my ethical obligation as a researcher, I am reporting that I have no financial or business interests in a company that may be affected by the research reported in the enclosed paper.

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